



Powder bed fusion process: A brief review

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ABSTRACT

Powder Bed Fusion (PBF) is an Additive Manufacturing (AM) process. It is the layer by layer addition process. In the future, it is going to be manufactured all types of products used in automotive, aerospace, energy sectors and home appliances. These products can be produced using various AM methods. Few of them are Laminated Object Manufacturing, Powder Bed Fusion and Direct Energy Deposition, Stereolithography (SLA), and Solid Ground Curing (SGC).

The present brief review is concentrated on working of PBF process only. It includes Direct Metal Laser Deposition (DMLS), Selective Laser Sintering (SLS), Selective Laser Melting (SLM), Electron Beam Melting (EBM) and Selective Heat Sintering (SHS). In these processes powder lay on the powder bed and it is also known as built platform. Laser used as heat energy source for SLS, SLM and DMLS processes. But EBM uses electron beam as source of energy. The SHS process uses heated thermal head for sintering plastic powders. SLS, DMLS and SHS are powder sintering processes. But SLM and EBM are powder melting processes.

This review paper is also covered historical development and usage of technologies, CNC verses Additive Manufacturing/Rapid Prototyping, AM process chain, Classification of AM processes; advantages, disadvantages and applications of PBF process.

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1. Introduction

1.1. Basic history

For fulfilling human needs, day by day new technologies have been developing. First basic historical development of technologies includes mechanization in the year 1770, invention of computer in 1946, Numerical Control (NC) machine tool in 1952, commercial LASER in 1960, commercial Robot in 1961, interactive graphics system in 1963, and Rapid Prototyping system in 1988 [1].

1.2. CNC vs. Additive Manufacturing/Rapid Prototyping (RP)

The difference between CNC systems and AM/RP systems is mention in the Table 1.

AM is a very quick production process compared to other processes. The production of end products based on traditional and advanced processes is given in the Table 2. Now Rapid Prototyping is merged with Additive Manufacturing.

Recently there is rapid increase in publications of metal AM, which led to the continuous progress. Based on the last 10 years

data available from Scopus database, there are approximately 2000 journal papers related to the topic of metal additive manufacturing [2] as shown in the Fig. 1. According to Wohler's Report 2011 (ISBN 0-9754429-6-1) the usage of AM in different fields is shown in Fig. 2.

1.3. Additive Manufacturing process chain

All Rapid Prototyping (RP) frameworks have a typical data work process. The RP process chain for the creation of 3D object is as per the following.

1. CAD model: All RP components can start from digital models, which are fully described by their geometries. It uses any CAD package.
2. Export data: The legitimate 3D display is sent out from the CAD bundle in a nonpartisan configuration, which as a rule is STL.
3. 3D model data repair: The sent out information is an estimation of exact inside 3D display. Amid this guess procedure the model surfaces are spoken with basic geometrical shapes as triangles. Tragically, STL models made along these lines can contain both-

Table 1
Difference between CNC systems and AM/RP systems.

	CNC systems	AM/RP systems
1	It is a material removal process.	It is a material adding process.
2	3D CAD models can be created in any suitable CAD packages.	3D CAD models can be obtained by modeling or digital scanning of a part or product and modified in the desired CAD packages.
3	It can accept IGES files, STEP files etc.	It can accept .STL, .SLC, AMF files etc.
4	3D CAD model data can be transferred to the desired CAM software.	First 3D CAD model data can be transferred to the desired slicing RP/AM software for slicing the CAD models and then generate g-code programs.
5	Part programmer can write CNC programs for tool path movements. Some CAM software's can generate CNC programs automatically.	Part programmer not required. All RP/AM software's can create CNC programs automatically for tool path movements.
6	Writing of part programs can take hours or days by the Part programmer.	Part programs can be generated in seconds or in minutes by the selected RP/AM software's.
7	Part programmer is responsible for writing correct programs.	The RP software is responsible for generating correct programs.
8	Program editing is possible and easier to identify editing blocks because of smaller length of the program.	Here also program editing is possible, but it takes more time for identifying editing blocks due to bigger length of the program.
9	Raw material cost is less or optimum.	Raw material cost is less or optimum for thermoplastic and more for metal powders.
10	Highly paid programmers and operators.	Highly paid engineers and less paid operators.
11	Time required to secondary operations on CNC parts is more.	Time required to secondary operations on AM parts is less.
12	Wastage of raw material is high.	Wastage of raw material is low.
13	Accuracy of parts is lesser /same as compared to AM parts.	Accuracy of parts is generally high.

ersome geometrical bundlers, for example, gaps and covering regions along the surface limits. Some RP bundles offers for repairs.

4. File manipulation: RP frameworks assemble components through the Z-axis of the STL models. Through reorientation of the parts with respect to the model organize frameworks; It enhances exactness, completeness and construction time.
5. Support structure generation: Fluid based Rapid Prototyping process forms the need to construct overhang territories of the parts. They are ordinarily created consequently utilizing particular programming apparatus.
6. Machine setup: Machine parameters are entered to determine the manufacturing way and wanted framework qualities. A 2D slice data generation is carried out.
7. Part building: Building of components is mostly an automated process by any AM system.
8. Removal of component and secondary operations: Parts can be removed and cleaned. Some components require extra operations before sending to the market.

Table 2
Basic difference among Traditional, RP and AM processes

	Product Design & Development	Manufacturing method	End Product Production time
Traditional process	Traditional	Traditional Manufacturing using conventional machines	Slower
Rapid Prototyping	Rapid	Traditional Manufacturing using NC, CNC, and DNC machines. Often, it is most probably Virtual Manufacturing (i.e. computer simulation)	Better, quick to supply parts to market and at less cost
Additive Manufacturing	Rapid	Additive Manufacturing using AM machines	Very fast to market, limited to AM applications

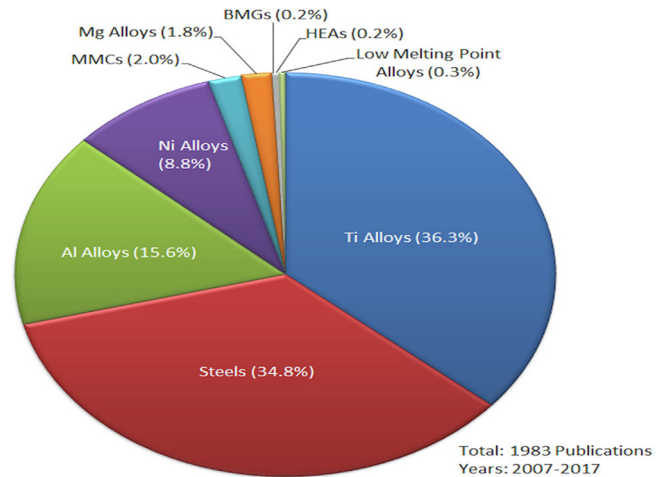


Fig. 1. The chart showing 10 years (2007–2017) of journal papers of various alloys used in AM () Source: www.scopus.com [2].

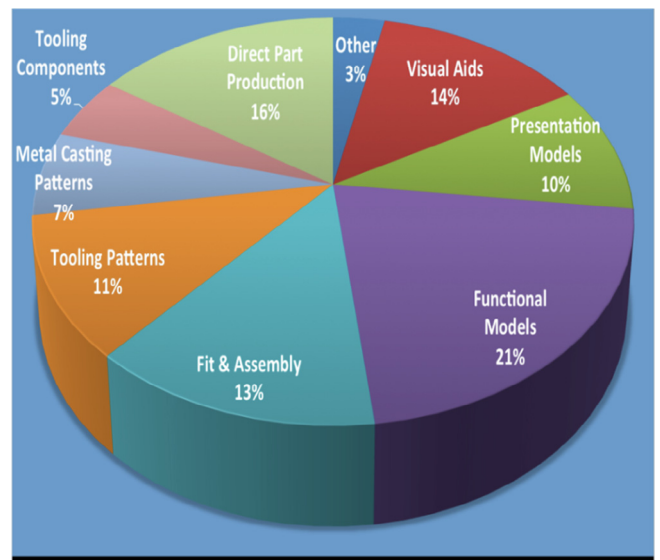


Fig. 2. Utilization of Additive Manufacturing for part building [Wohler's Report 2011].

1.4. Categories of AM methods

All liquid, solid and powder based AM methods has categorized by the ASTM F42 committee into seven broad areas [3,4,5,6] which are shown in Fig. 3 and table 3.

Details of Powder Bed AM methods/systems researchers, research industries/institutions, related patents and present AM systems producers are given in the table4 and table5.

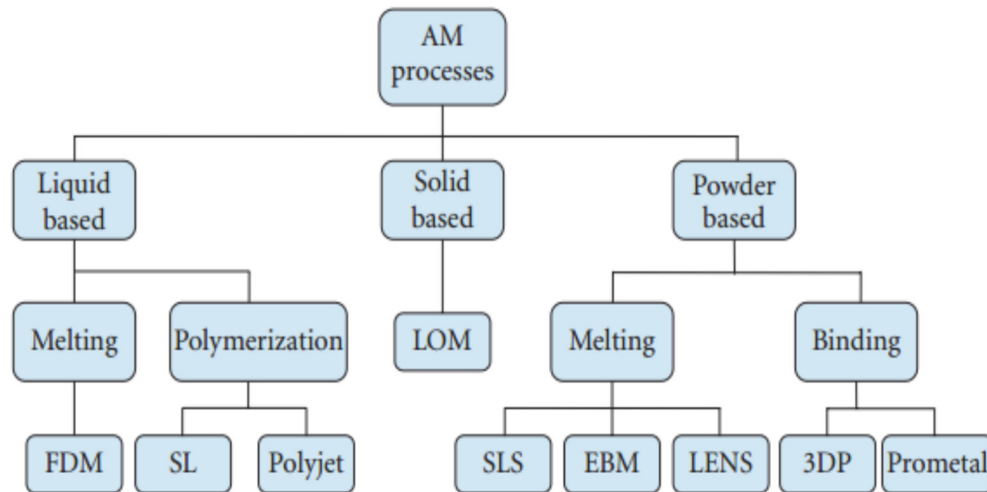


Fig. 3. Classification of AM processes [3,4,5,6].

Table 3

Categorizing of various AM processes (Ian Gibson *et al.*, 2010; Dr. Wei Jun 2013) [3,4,6].

Process Type	Related Technology	Material Used
Vat Stereolithography (SLA), digital light processing (DLP)	Photopolymerization Photopolymers	
Material Jetting	Multi-jet modeling (MJM)	Polymers, Waxes
Binder Jetting	Powder bed and inkjet head (PBIH), plaster-based 3D printing (PP)	Polymers, Foundry Sand, Metals
Material Extrusion Powder bed fusion	Fused deposition modeling (FDM) Electron beam melting (EBM), selective laser sintering (SLS), selective heat sintering (SHS), and direct metal laser sintering (DMLS)	Polymers Metals, Polymers
Sheet Lamination	Laminated object manufacturing (LOM), ultrasonic consolidation (UC)	Paper, Metals
Directed Energy Deposition	Laser metal deposition (LMD)	Metals

1.5. Advantages, disadvantages and applications

1.5.1. Advantages

1. RP system can allow the development of products in three weeks since 1998.
2. Use of many structural powders and polymers in SLS.
3. Support structures are not required in SLS process.
4. SLS parts are cheaper than EBM parts.
5. EBM is one of the method for production high-quality metal components with free of internal stresses using wide range of metal powders.
6. Fully dense parts with very homogeneous microstructures can be obtained by EBM. But SLM produces only dense parts and may be homogeneous microstructures.
7. Microstructural control is better in EBM compared to conventional processes.
8. Alloy components can be made by DMLS with full melting of metal powders.
9. The above mention AM processes can manufacture complex parts with little effort.
10. PBF process reduces the raw materials stock and wastage, material transport costs, inventory.

Table 4

Details of laser sintering and laser melting [1].

S. No	Process	SLS	DMLS	SLM	Selective Laser reMelting (SLrM)
1	Founded (in year)	1987	----	2000	----
2	First granted U.S patent (in year)	4,863,539(1986)	----	19,649,865(2000)	----
3	Inventor	Carl Deckard, Joe Beaman	----	First successfully demonstration at Fraunhofer ILT in 2008	----
4	Company	University of Texas; Later DTM corp.	EOS	EOS	University of Liver pool, Later merged with EOS GmbH, Germany
5	Head Quarter	Austin (U.S.A)	Germany	Europe	Liver pool (U.K)
6	First system introduced	1992	1995	2008	----
7	Feed stock	Thermoplastic and powders of metal	Powders metal alloys	Powders metal alloys	Powders metal alloys
8	Exist	2011: Acquired by 3D systems	----	----	----
9	Initial target	Industry	Industry	Industry	Industry
10	Current manufacturer	3D systems	EOSINT, Germany	EOS GmbH, Germany	EOS GmbH, Germany

Table 5
Details of SHS and EBM [1].

S. No	Process	SHS	Two organizations of EBM	
			EBM	EBM
1	Founded (in year)	-----	1993	1997
2	First granted U.S patent (in year)	-----	5,786,562(1993)	6,519,500(1997)
3	Inventor	Blue Printer SHS	Ralf Larson	Chalmers University of Technology,Sweden
4	Company	-----	Larson Brothers	ARCAM
5	Head Quarter	2011	-----	Sweden, Cambridge (MA)
6	First system introduced	-----	-----	2002
7	Feed stock	Powders of thermoplastic	Powders metal alloys	Metal Powders
8	Exist	-----	-----	2000: IPO (Nordic Growth Market)
9	Initial target	-----	Commercialization	Industry
10	Current manufacturer	-----	-----	ARCAM

11. Complex and accurate dimensional parts can be made by PBF process from different material powders.
12. Desired mechanical PBF parts can be obtained by selecting required parameters.

1.5.2. Disadvantages

1. SLS process is an expensive relative to Fused Deposition Modeling and 3Dimensiona Printing.
2. SLS metal parts quality is lower than EBM parts.
3. EBM requires vacuum operation and it is an expensive.
4. Traditional secondary operations are needed for EBM parts.
5. Cost of PBF equipment's and materials is high in case of powder materials usage.
6. SHS process can use only plastic powders as raw materials.

1.5.3. Applications

1. Prototypes for demonstrational purpose.
2. Conceptual Models, Engineering Models, and Functional Prototypes.
3. Production of structural components by SLS, EBM
4. Custom-made bio-implants (Ti6Al4V) using EBM and SLM.
5. Applications in mold making and Tooling.
6. Applications in Aerospace, automotive industries.

2. Powder Bed fusion (PBF) process

PBF process is an advanced production technology that can be utilized to fabricate actual parts from the desired 3D solid CAD model data based on layer by layer addition concept [1,2,3,6,7,8,9,10,11,12]. This process uses either laser or electron beam as the heat energy sources for irradiation, fusion and melting of powder particles. The most commonly used heat sources in PBF are various types of lasers. The types of PBF processes are

1. Selective Laser Sintering (SLS)
2. Selective Laser Melting (SLM)
3. Direct Metal Laser Sintering (DMLS)
4. Selective Heat Sintering (SHS)
5. Electron Beam Melting (EBM)

Sintering process by laser, again classified based as polymer Laser Sintering (pLS) process and metal Laser Sintering (mLS) process. These processes were carried out by pLS machine and mLS machine. Electron Beam Melting is used to create parts with metals and alloys powders in a vacuum built chamber. Almost all PBF methods involved spreading of powders over previous layers. This can be done by a roller or a blade mechanism. PBF techniques [13,14] start from the spreading of powders and the beam scans

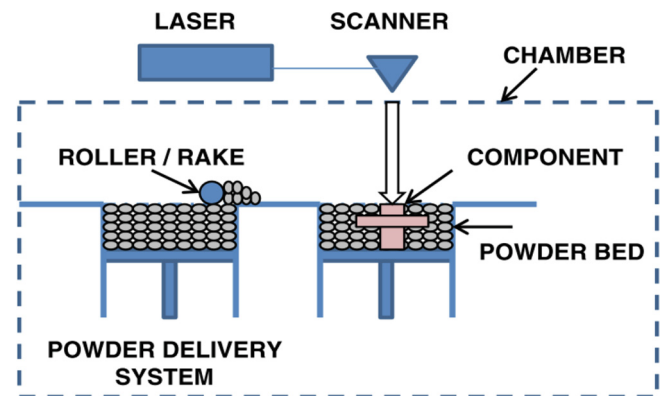


Fig. 4. Selective Laser Sintering [13].

the powders that fuses the powders and solidifies it as end components. These methods manufacture accurate and good surface finish parts for small scale production because of smaller beam size.

2.1. Selective Laser Sintering (SLS) process

The illustration of SLS method with respect to the Fig. 4 is given here. Powder can be spread on the built platform by roller mechanism, when the powder delivery piston moves up. The laser beam can scan the powders selectively and fuse/sinter the powder particles based on 3D CAD data, forming the first sintered layer. Part building bed can move down by the required thickness and second layer of powder is spread on first sintered layer. Then laser beam again scans selectively and forms the second layer by fusion onto the first layer. The process can repeat till the complete building of physical part. The part is taken from the RP machine cleaning and for secondary processes if required [1,2,3,6,7,8,9,10,11,12,15]. Sometimes CO₂ laser beam is used [16].

In metal SLS process a mixture powder is used. It consists of one powder having low melting point and another high melting point. During SLS process only low melting point powders can partially melt and forms the bond with unmelt high melting point powders.

2.2. Selective Laser melting (SLM) process

The working of SLM process is similar to that of SLS process. It melts alloy powder particles instead of sintering [1,2,6,7,8,9,10,12,15,17,18]. It is shown in Fig. 5. The manufacturing cycle time of complex part production is rapidly reduced. Finer microstructure can be observed at high cooling rate compared to conventional manufacturing methods. SLM parts have excellent mechanical properties. Aerospace, automobile, and medical applications parts are produced by SLM. Lattice structure can be pro-

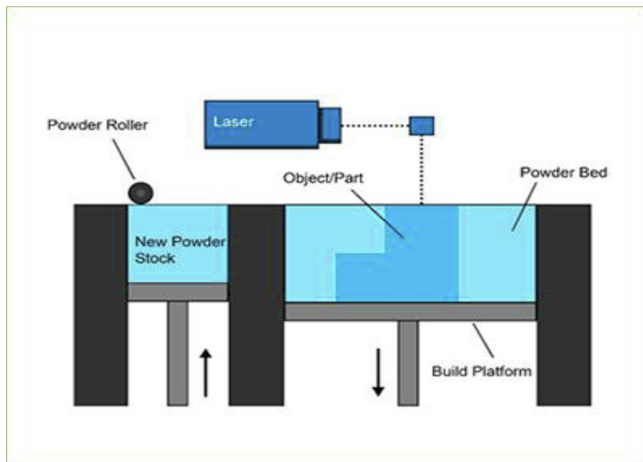


Fig. 5. Selective laser melting [9].

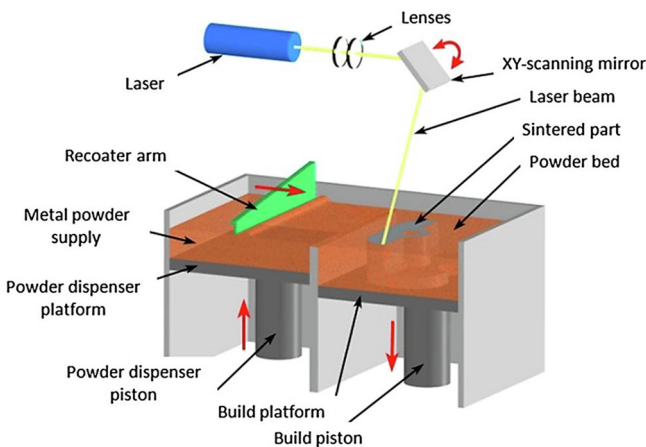


Fig. 6. Direct Metal Laser Sintering (DMLS) Setup [14].

duced using Selective Laser Melting [19]. Defects can be identified and evaluated [20].

2.3. Direct metal Laser Sintering (DMLS) process

Fig. 6 showing the DMLS process [1,2,7,8,9,10,12], which is similar to that of SLM process. Functional prototypes, short-run components, and functionally graded materials (FGM) are manufactured by direct metal laser sintering. Aerospace parts can be produced using less raw materials and with low cost.

2.4. Selective heat Sintering (SHS) process

Plastic powder particles fuse together by a heated head. The heated head touches the powder and moves based on sliced STL model. This method is used for manufacturing structural parts and conceptual prototypes. The SHS technology is used in Blue printer and it is a desktop 3D printer [9]. It has a built chamber of size 200mmx160mmx140mm, printing speed of 0.078–0.118 in./hour with a layer thickness of 0.0039 in..

2.5. Electron beam melting (EBM) process

Electron Beam Melting (EBM) is an another powder fusion process uses beam of electrons as heat energy source for melting metal powders [1,2,3,6,7,9,11,12,17] as shown in Fig. 7. In EBM,

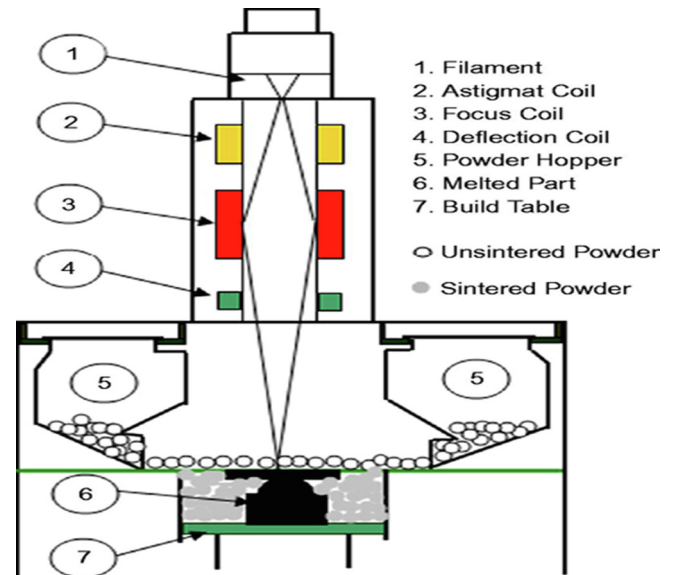


Fig. 7. Electron Beam Melting (EBM) [Wohler's Report 2011].

electrons are emitted from a heated tungsten filament at a high speed, which can be controlled by two magnetic fields, focusing coil and deflecting coil. The focusing coil creates a thin diameter of 0.1 mm electron beam and deflecting coil deflect the focused beam at desired points for scanning the layer of alloy powder.

The working of EBM is as follows. High vacuum and high temperature can be maintained in the built chamber. Powder hoppers pour a layer of alloy powders on built chamber. The powder particles melt in a small volume within the layer by a focused electron beam. The electron beam is scanned to define a 2D slice of the object within the layer. The built chamber can be moved down, and a second layer of alloy powder is poured onto the top of the previous solidified layer. The process repeats till completion of part. After removal of part from the machine, unmelt metal powders are cleaned and it can be recycled for further use. Corrosive resistance Hastelloy X can be processed by EBM and also by SLM [21].

3. Conclusions

From this review article the following conclusions are drawn.

1. Here, the working processes of SLS, SLM, DMLS, SHS and EBM are described. These processes are part of PBF process.
2. PBF process used for producing precise complex parts and products. More précised objects can be fabricated using SLM and EBM.
3. Physical objects obtained from the processes of powder sintering such as DMLS, SLS, and SHS had less density, than powder melting processes (i.e. SLM and EBM).
4. This article is also discussed technological development, CNC verses AM/RP systems, classification of AM processes; advantage, disadvantages and applications of PBF process.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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